Search Without a Utility Function

Can we evaluate a newly invented game before we know how to play it well?

Research proposal · UCSC Augmented Design Lab

Two different questions

I(G) = strategic interestingness

F(G) = player appeal / fun

Neither is simply the game’s win/loss payoff.

Can these quantities be measured independently?

What should I(G) capture?

DECISION SIGNIFICANCE

$D(G) = E_S[\max_a V(s,a) - \min_a V(s,a)]$

SKILL SENSITIVITY

$S(G) = P(\text{stronger agent wins} \mid G)$

STRATEGIC DIVERSITY

Are several distinct plans viable?

V(s,a): estimated outcome after action a under a specified continuation policy.
Candidate measures—not yet a definition of I(G).

How might we estimate F(G)?

Goal: predict player appeal from a game's rules—not require a new user study for each game.

Input .lud rules → agent self-play and diagnostics → predictive model $\hat{F}(G)$.

Train and calibrate using historical games and existing preference data.

Validate on held-out games, historical adoption, and optional human studies.

Test: could we predict which historical chess variants persisted?

Ratings are instruments, not answers

ELO · Arpad Elo · US Chess 1960; FIDE 1970
$R' = R + K(S - E)$

GLICKO-2 · Mark E. Glickman · 2000
Player = (r, RD, σ)
Rating · uncertainty · volatility

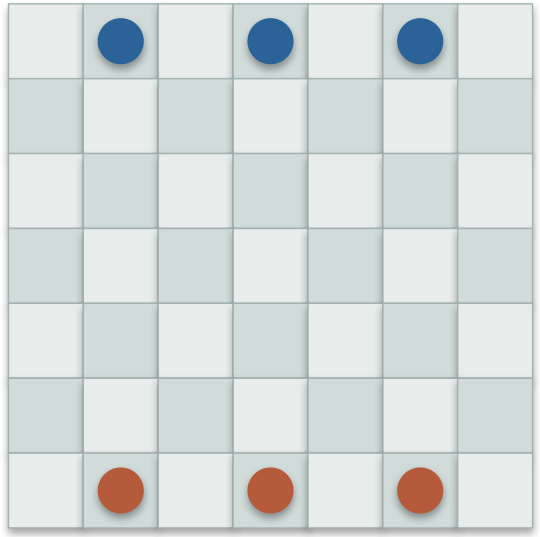
BRADLEY–TERRY · pairwise preferences

Historical arc: Elo predicts performance; Glicko-2 models uncertainty, inactivity and changing strength.
Glickman's 1993 Harvard PhD helped lead to Glicko. Ratings support I(G); they do not define it.

First test case: Ember Roads

Validate rules → sample play → measure I(G) → predict F(G)

7×7 board; three pawns per side; orthogonal movement leaves burned squares.



Ember Roads · schematic

Established: Ludii compiles; 9 initial one-step moves; turns alternate; 20/20 random games terminated.

Discrepancy: written rules imply 3 additional two-step moves.

Not yet verified: winner correctness and ember identity.

Research questions and next steps

Can we distinguish interesting games from trivial ones?

Can strategic interest be high when fun is low?

Can the pipeline work across Ludii and other engines?

Next: finish rules validation; specify measurable I components; design a small human-preference pilot.

No claim yet that I(G) or F(G) has been established.

References — foundations for I(G) and F(G)

Lantz et al. (2017) · Depth in Strategic Games

Formalizes strategic depth through improvement with increasing computation.

https://www.nealen.net/papers/Lantz2017Depth.pdf

Browne & Maire (2010) · Evolutionary Game Design

Self-play metrics for evaluating and generating combinatorial games.

https://doi.org/10.1109/TCIAIG.2010.2041928

Yannakakis & Hallam (2007) · Towards Optimizing Entertainment

Quantitative entertainment features tested against human judgments.

https://doi.org/10.1080/08839510701527580

Bradley & Terry (1952) · Method of Paired Comparisons

Statistical model for estimating relative human preferences.

https://doi.org/10.1093/biomet/39.3-4.324

Glickman (1999) · Dynamic Paired Comparison Experiments

Tracks changing skill while accounting for rating uncertainty.

https://doi.org/10.1111/1467-9876.00159

Our proposal: measure I(G) and population-specific F(G) separately, then test how they relate.